
Conditioning Stochastic Processes using Guided Latent Dynamics

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Abstract

Stochastic-process models are, as a rule, far easier to simulate than to condition. Non-linear observations, non-Gaussian likelihoods, black-box information, and global constraints all induce intractable conditional laws, requiring bespoke, model-specific constructions. We introduce LATENTFLOW, a single framework for conditioning stochastic processes, with no learned neural approximations and no training. Our starting point is to write the stochastic process as the deterministic image of a tractable latent innovation, $f_0 = T_\vartheta(\xi_0)$, with ξ_0 sampled from a simple reference distribution. This reduces process-level conditioning to latent-space inference: pull the likelihood back through T_ϑ , sample the resulting latent law with a tractable guided probability flow, and push the samples forward. This construction is provably exact at the level of the target law; in practice, approximation enters only through finite terminal noising, Monte Carlo guidance, and time discretisation of the continuous-time dynamics, each of which is explicit and systematically reducible. As LATENTFLOW is training-free, conditioning reduces to solving a single reverse-time SDE. This enables conditional sampling in seconds on a single desktop CPU across model classes that have never shared a scalable method: classical spatial priors, nonlinear stochastic dynamics, mechanistic models from the physical and life sciences, stochastic PDEs, heavy-tails and extremes, point and discrete-state processes, and neural or simulator-defined processes.

1 Introduction

Sampling from a stochastic process only requires running it forward; conditioning it, until now, has in many cases remained the preserve of bespoke, model-specific constructions. Examples include diffusion paths conditioned on fixed endpoints or partial observations [Delyon and Hu, 2006, Schauer et al., 2017]; functions with shape [Riihimäki and Vehtari, 2010] or physics-informed [Chen et al., 2021, Hamelijncx et al., 2024] constraints; and simulator outputs conditioned on indirect measurements, rare events, or user-defined criteria [Botev and L’Ecuyer, 2020, Finzi et al., 2023]. In each case the conditioning information cannot be absorbed into the model in closed form, demanding its own specialised, expensive, or approximate sampler; for instance, those based on MCMC [Andrieu et al., 2003], SMC [Doucet et al., 2001], variational inference [Blei et al., 2017] or Laplace approximations [Rue et al., 2009].

Recently, Moss et al. [2026] introduced FLOWGP, which addresses conditioning for Gaussian process (GP) priors under intractable information by recasting conditional sampling as a differential equation with closed-form Gaussian dynamics and a likelihood-dependent guidance term. The resulting sampler applies to any likelihood that can be evaluated pointwise, and requires no trained network to approximate the guidance field. Underlying the model is a simple mechanism: the GP sample is the linear, deterministic image of a Gaussian innovation, and conditioning is a re-weighting of that innovation’s density by the pulled-back likelihood.

Our key observation is that Gaussianity of the stochastic process prior is not the essential ingredient in this construction; instead, it is the availability of a tractable *latent innovation representation* on which guided dynamics can be defined. Rather than constructing conditioning dynamics directly in the original process space, this allows us to express the process as the deterministic image of a tractable innovation variable, pull the conditioning information back through this generator, and sample the resulting reweighted law in latent space. The conditioned process is then obtained by pushing the guided samples in the latent space forward through the same generator. We refer to this method as LATENTFLOW. Our construction does not require learning a process prior, score model, or conditional generator [e.g., Song et al., 2021, Chung et al., 2023, Zammit-Mangion et al., 2025]; the sampler is built directly from the known map T_θ and pointwise evaluations of the pulled-back likelihood. Our method also provides a natural way to learn the parameters of the process generator.

We obtain formal theoretical guarantees for LATENTFLOW, and demonstrate its application to a wide variety of stochastic process priors. These include GPs; heavy-tailed Student- t and Cauchy-convolution processes; discrete-valued Potts processes; linear and nonlinear diffusion processes including a model for cell-differentiation, a FitzHugh–Nagumo model for excitable neurons, and a Heston stochastic volatility model; mechanistic models of SIR epidemics and Lotka–Volterra predator–prey dynamics; and Allen–Cahn and advection–diffusion SPDEs. See Figure 1 for several examples. Until now, beyond simple or contrived cases, conditioning each of these processes required either low-fidelity approximations or expensive computation. We sample in single-seconds time on a single desktop CPU, providing the first general method for conditioning stochastic processes in real time which is exact up to an explicit and systematically reducible numerical error.

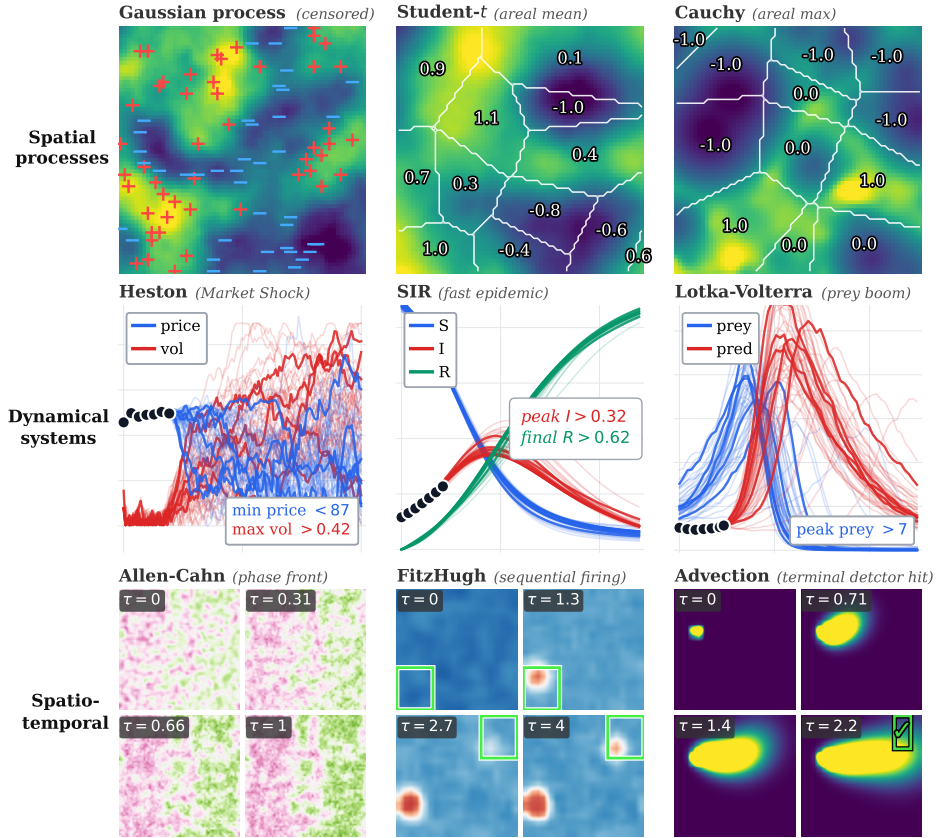


Figure 1: LATENTFLOW across diverse stochastic-process priors and conditions. **Top:** spatial processes conditioned on censored point measurements for a Gaussian process; prescribed areal means for a Student- t process; and prescribed areal maxima for a Cauchy convolution field. **Centre:** SDEs conditioned on observations and events. A Heston model under a joint market-crash and volatility spike; an SIR model under a fast epidemic peak; and a Lotka–Volterra system given a large prey boom. **Bottom:** Spatio-temporal PDEs. Allen–Cahn conditioned on a prescribed phase front (negative left, positive right) at final time; FitzHugh–Nagumo conditioned on sequential source firing (bottom-left then top-right); and an advection–diffusion plume conditioned on reaching a downstream detector (box).

2 Conditioning via a latent representation

Let $\ell \in \mathcal{E}$ denote a stochastic process, viewed as a random element in a measurable space $\mathcal{E}$, and let $f_0 \in E_m$ denote a finite-dimensional discretisation of ℓ . Let $p_\vartheta(f_0)$ denote the prior density of f_0 on E_m , with hyperparameters ϑ . We are interested in sampling the conditional distribution of this process, given some additional information $\mathcal{C}$ encoded by a likelihood $L_{\mathcal{C}} : E_m \rightarrow [0, \infty)$. The desired conditional density is

$$\pi_\vartheta(f_0 | \mathcal{C}) = \frac{L_{\mathcal{C}}(f_0)p_\vartheta(f_0)}{Z_\vartheta}, \quad Z_\vartheta = \int_{E_m} L_{\mathcal{C}}(f_0)p_\vartheta(f_0) \, df_0.$$

Our key assumption is that p_ϑ can be obtained by transforming a simple reference density.

Assumption 2.1. There exists a latent dimension m_ξ , a reference density r on $\mathbb{R}^{m_\xi}$, and a map $T_\vartheta : \mathbb{R}^{m_\xi} \rightarrow E_m$ such that, if $\xi_0 \sim r$, then $f_0 = T_\vartheta(\xi_0)$ has density p_ϑ .

After discretisation, this holds for essentially any process of interest; representative innovation maps are given in [Section A.1](#). We fix $r(\xi_0) = \mathcal{N}(\xi_0; 0, I_{m_\xi})$ throughout.

Once such a representation is available, conditioning the process on $\mathcal{C}$ can be transferred to the latent space via the *latent likelihood* $\Lambda_{\vartheta, \mathcal{C}}(\xi_0) = (L_{\mathcal{C}} \circ T_\vartheta)(\xi_0)$. The corresponding latent target density is

$$\rho_\vartheta(\xi_0 | \mathcal{C}) = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_0) r(\xi_0)}{Z_\vartheta}, \quad Z_\vartheta = \int_{\mathbb{R}^{m_\xi}} \Lambda_{\vartheta, \mathcal{C}}(\xi_0) r(\xi_0) \, d\xi_0,$$

where the posterior is recovered by pushing ρ_ϑ through T_ϑ . That is, if $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$, then $f_0 = T_\vartheta(\xi_0)$ has density $\pi_\vartheta(\cdot | \mathcal{C})$. We formalise this in the following proposition.

Proposition 2.2. Suppose that [Assumption 2.1](#) holds and that $0 < Z_\vartheta < \infty$. Then, if $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$, the random variable $f_0 = T_\vartheta(\xi_0)$ has the conditioned process density $\pi_\vartheta(\cdot | \mathcal{C})$.

3 LATENTFLOW: inference with guided latent flows

If the latent likelihood is differentiable, or its score admits a tractable estimator, we can use a guided diffusion to draw samples $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$. We call the resulting framework LATENTFLOW.

Let $\xi_0 \sim r$ and $\varepsilon \sim \mathcal{N}(0, I_{m_\xi})$ be independent of ξ_0 . The variance-preserving (VP) latent noising bridge is given by

$$\xi_t = \alpha(t)\xi_0 + \sigma(t)\varepsilon, \tag{1}$$

where $\alpha(t) = \exp\{-\frac{1}{2} \int_0^t \beta(s) \, ds\}$ and $\sigma(t)^2 = 1 - \alpha(t)^2$, for $\beta : [0, 1) \rightarrow (0, \infty)$ such that $\lim_{t \uparrow 1} \alpha(t) = 0$. Since $r(\xi_0) = \mathcal{N}(\xi_0; 0, I_{m_\xi})$, we have $\xi_t \sim r$ for every $t \in [0, 1)$.

The additional condition enters through the latent likelihood Λ . For $t \in [0, 1)$, we define the *latent conditioning potential* as

$$\psi(t, u) = \mathbb{E}_{\xi_0 \sim r} [\Lambda_{\vartheta, \mathcal{C}}(\xi_0) | \xi_t = u] = \mathbb{E}_{\varepsilon \sim r} [\Lambda_{\vartheta, \mathcal{C}}(\alpha(t)u + \sigma(t)\varepsilon)]. \tag{2}$$

This can be viewed as a smoothed version of the latent likelihood. Finally, suppose $\psi(t, u) > 0$ and that $\psi(t, \cdot)$ is differentiable at u . We define for $t \in [0, 1)$ the *latent guidance field*

$$g(t, u) = \nabla_u \log \psi(t, u)$$

Our goal is a sampling procedure for $\xi_0 \sim \rho_0$ with $\rho_0 \equiv \rho_\vartheta$. Let $\rho_t = \mathcal{L}(\xi_t)$ denote the law of the random variable ξ_t obtained by drawing $\xi_0 \sim \rho_0$ and then applying the noising bridge in (1). The family of guided densities $(\rho_t)_{0 \leq t < 1}$ are the one-time marginals of the VP-SDE

$$d\xi_t = -\frac{1}{2}\beta(t)\xi_t \, dt + \sqrt{\beta(t)} \, dw_t, \quad \xi_0 \sim \rho_0, \quad t \in [0, 1),$$

where w is a standard $\mathbb{R}^{m_\xi}$ -valued Brownian motion [e.g., [Song et al., 2021](#)]. By [Anderson \[1982\]](#), the time reversal of these dynamics is also an SDE, given by

$$d\xi_t = \beta(t) \left[\frac{1}{2}\xi_t - g(t, \xi_t) \right] dt + \sqrt{\beta(t)} \, d\bar{w}_t, \tag{3}$$

where $\bar{w}$ is a reverse-time Brownian motion, and this equation is integrated backwards in t . In particular, if (3) is initialised at $\xi_{t_1} \sim \rho_{t_1}$ and integrated backwards from t_1 to 0, then $\xi_t \sim \rho_t$ for

every $t \in [0, t_1]$, and hence $\xi_0 \sim \rho_0$. Since $\rho_{t_1} \Rightarrow \mathcal{N}(0, I_{m_\xi})$ as $t_1 \uparrow 1$, initialising at $\mathcal{N}(0, I_{m_\xi})$ in place of ρ_{t_1} is asymptotically exact, yielding a practical sampler.

In general neither $g(t, u)$ nor $\psi(t, u)$ are available in closed form. We can, however, obtain estimates of both via a Monte Carlo estimator of (2). For a given pair (t, u) , draw $\varepsilon^{(s)} \sim \mathcal{N}(0, I_{m_\xi})$, and set $\xi_{0|t}^{(s)} = \alpha(t)u + \sigma(t)\varepsilon^{(s)}$, for $s = 1, \dots, S$. The self-normalised estimator is given by

$$\hat{g}_S(t, u) = \alpha(t) \sum_{s=1}^S \bar{w}_s \nabla_\xi \log \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(s)}), \quad \bar{w}_s = \frac{\Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(s)})}{\sum_{\ell=1}^S \Lambda_{\vartheta, \mathcal{C}}(\xi_{0|t}^{(\ell)})}.$$

Since $\Lambda_{\vartheta, \mathcal{C}} = L_{\mathcal{C}} \circ T_{\vartheta}$, the latent score can be computed by automatic differentiation through the process generator. In practice, we replace $g(t, u)$ by $\hat{g}_S(t, u)$ and solve (3) numerically. The resulting error decomposes into three terms: terminal initialisation, guidance approximation, and numerical discretisation; see Sharrock et al. [2026] for further details. Each of these can be made arbitrarily small, so the sampling error is limited only by available compute. These bounds pass to process space whenever T_{ϑ} is Lipschitz.

4 Numerical experiments

Below we provide an illustrative application of LATENTFLOW: conditioning spatio-temporal models. Additional plots, as well as full implementation details, are provided in Appendix B. Such models arise in turbulent transport, environmental dispersion, materials science, excitable media, and phase separation [Majda and Kramer, 1999, Allen and Cahn, 1979, Nagumo et al., 1962]. Conditioning such models on sparse observations, future events, or aggregate space–time constraints is central to data assimilation, inverse modelling, and rare-event prediction.

We consider three examples: a stochastic Allen–Cahn field for noisy phase separation, an advection–diffusion plume model, and a stochastic FitzHugh–Nagumo excitable-media model that produces travelling activation waves [FitzHugh, 1961, Nagumo et al., 1962, Allen and Cahn, 1979, Majda and Kramer, 1999]. Representative samples are shown in Figure 2. Across all three models, LATENTFLOW produces fields that remain consistent with the prior dynamics while satisfying the imposed conditions.

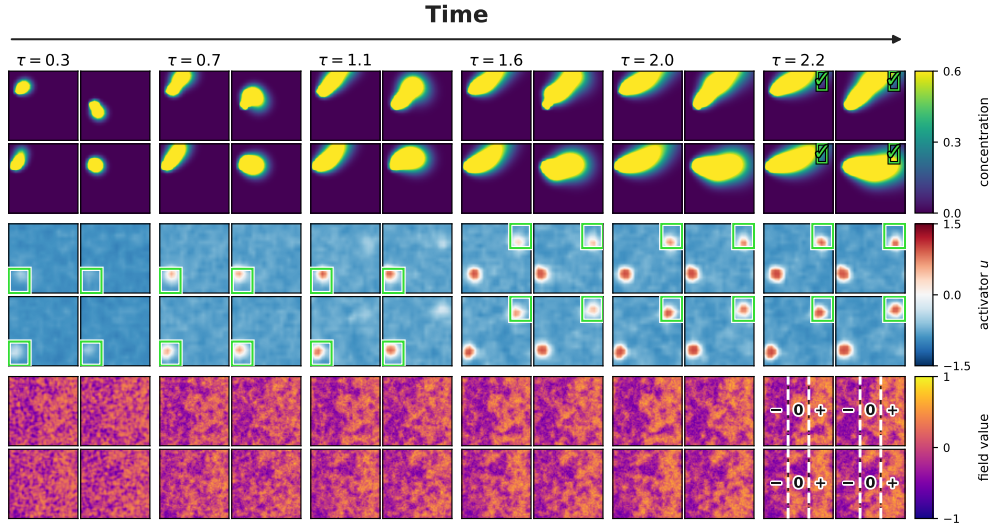


Figure 2: Posterior samples from three spatio-temporal systems using our LATENTFLOW: **(top)** a concentration plume conditioned on triggering a downstream detector evolving under advection–diffusion dynamics (160 latent dimensions); **(middle)** a FitzHugh–Nagumo excitable medium conditioned on sequential pacemaker firing (40K latent dimensions); and **(bottom)** an Allen–Cahn phase field conditioned on a prescribed phase front (122K latent dimensions). Each row shows four independent draws, over six time snapshots. Sampling time: 1s for the advection–diffusion; 5s for FitzHugh–Nagumo, and 2s for Allen–Cahn.

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A Additional technical details

A.1 Examples of latent innovation representations

Below we provide three representative stochastic processes that satisfy [Assumption 2.1](#).

Gaussian process. Let $f \sim \text{GP}(\mu_\vartheta, k_\vartheta)$, and let $f_0 = (f(s_1), \dots, f(s_m)) \in \mathbb{R}^m$ denote its values at inputs $s_{1:m}$. Thus $f_0 \sim \mathcal{N}(m_\vartheta, K_\vartheta)$, where $m_\vartheta = (\mu_\vartheta(s_1), \dots, \mu_\vartheta(s_m))$ and $(K_\vartheta)_{ij} = k_\vartheta(s_i, s_j)$. Let L_ϑ satisfy $L_\vartheta L_\vartheta^\top = K_\vartheta$. We can then write

$$f_0 = T_\vartheta(\xi_0) = m_\vartheta + L_\vartheta \xi_0, \quad \xi_0 \sim \mathcal{N}(0, I_m).$$

Thus, the latent innovation is the whitened Gaussian vector ξ_0 , and the generator T_ϑ is the usual affine GP whitening map.

Student- t process. Let $f \sim \text{TP}_\nu(\mu_\vartheta, k_\vartheta)$, where k_ϑ denotes the Student- t scale kernel, and let $f_0 = (f(s_1), \dots, f(s_m))$. We then have

$$f_0 = m_\vartheta + \omega^{-1/2} L_\vartheta v,$$

where $v \sim \mathcal{N}(0, I_m)$ and $\omega \sim \text{Gamma}(\nu/2, \nu/2)$ independently. Let $a \sim \mathcal{N}(0, 1)$, and define $G_\nu(a) = F_{\Gamma, \nu}^{-1}(\Phi_N(a))$, where $F_{\Gamma, \nu}$ is the $\text{Gamma}(\nu/2, \nu/2)$ CDF and Φ_N is the standard normal CDF. Then

$$f_0 = T_\vartheta(\xi_0) = m_\vartheta + G_\nu(a)^{-1/2} L_\vartheta v, \quad \xi_0 = (v, a) \sim \mathcal{N}(0, I_{m+1}).$$

Thus, m Gaussian variables generate the spatial Gaussian component, and one additional Gaussian variable generates the random scale.

SDE-driven process. Consider a process defined via the stochastic differential equation (SDE)

$$dx_\tau = b_\vartheta(\tau, x_\tau) d\tau + \Sigma_\vartheta(\tau, x_\tau) dw_\tau.$$

On a grid $0 = \tau_0 < \tau_1 < \dots < \tau_N = T$, the Euler–Maruyama discretisation is

$$x_{n+1} = x_n + b_\vartheta(\tau_n, x_n) \Delta_n + \Sigma_\vartheta(\tau_n, x_n) \sqrt{\Delta_n} \eta_n, \quad \eta_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I).$$

If x_0 is random, we also write $x_0 = \chi_\vartheta(\eta_{-1})$, with $\eta_{-1} \sim \mathcal{N}(0, I)$, and omit η_{-1} when x_0 is fixed. The latent innovation is $\xi_0 = (\eta_{-1}, \eta_0, \dots, \eta_{N-1})$, while the generator T_ϑ is the numerical solver itself: it applies the recursion and returns the discretised path $(x_0, x_1, \dots, x_N) = T_\vartheta(\xi_0)$.

A.2 Proof of the latent conditioning principle

Proof of [Proposition 2.2](#). Let $\xi_0 \sim \rho_\vartheta(\cdot | \mathcal{C})$. Then, for any measurable $A \subseteq E_m$,

$$\begin{aligned} \mathbb{P}(T_\vartheta(\xi_0) \in A) &= \frac{1}{Z_\vartheta} \int_{\mathbb{R}^{m_\xi}} \mathbf{1}_{\{T_\vartheta(\xi_0) \in A\}} L_{\mathcal{C}}(T_\vartheta(\xi_0)) r(\xi_0) d\xi_0 \\ &= \frac{1}{Z_\vartheta} \int_A L_{\mathcal{C}}(f_0) p_\vartheta(f_0) df_0 = \int_A \pi_\vartheta(f_0 | \mathcal{C}) df_0. \end{aligned}$$

The second equality follows from the pushforward relation $T_{\vartheta\#}(r(\xi_0) d\xi_0) = p_\vartheta(f_0) df_0$, applied to the function $g(f_0) = \mathbf{1}_A(f_0) L_{\mathcal{C}}(f_0)$. It follows immediately that $T_\vartheta(\xi_0)$ has density $\pi_\vartheta(\cdot | \mathcal{C})$. $\square$

A.3 Exactness of the latent dynamics

Fix ϑ and $\mathcal{C}$, and write $\Lambda = \Lambda_{\vartheta, \mathcal{C}}$, $Z = Z_{\vartheta}$, and $\rho_0 = \rho_{\vartheta}(\cdot \mid \mathcal{C})$. For $t \in [0, 1]$, let ρ_t denote the density obtained by noising ρ_0 through (1); equivalently,

$$\rho_t(u) = \frac{\psi(t, u)r(u)}{Z}.$$

Assumption A.1. For every compact interval $[t_0, t_1] \subset (0, 1)$, ρ_t is a strictly positive Lebesgue density and $(t, u) \mapsto \nabla_u \log \rho_t(u)$ is sufficiently regular that the reverse-time SDE below is well posed and its Fokker–Planck equation has a unique density-valued solution.

Theorem A.2. Suppose that [Assumptions A.1](#) and [2.1](#) hold, that $r = \mathcal{N}(0, I_{m_\epsilon})$, and that $0 < Z < \infty$. Fix $0 < t_0 < t_1 < 1$, define $\tau_s = t_1 - s$, and let

$$d\xi_s^\leftarrow = [\beta(\tau_s)g(\tau_s, \xi_s^\leftarrow) - \tfrac{1}{2}\beta(\tau_s)\xi_s^\leftarrow] ds + \sqrt{\beta(\tau_s)} d\bar{w}_s, \quad \xi_0^\leftarrow \sim \rho_{t_1}. \quad (4)$$

Then $\xi_s^\leftarrow \sim \rho_{\tau_s}$ for all $0 \leq s \leq t_1 - t_0$. If the reverse SDE is well defined down to $t_0 = 0$, then $\xi_{t_1}^\leftarrow \sim \rho_0$ and $T_{\vartheta}(\xi_{t_1}^\leftarrow) \sim \pi_{\vartheta}(\cdot \mid \mathcal{C})$.

Proof. Let $b_t(u) = -\frac{1}{2}\beta(t)u$ denote the drift of the VP SDE. By the standard time-reversal formula for diffusions [e.g., [Anderson, 1982](#)], the reverse-time diffusion with clock $\tau_s = t_1 - s$ has drift

$$\tilde{b}_s(u) = -b_{\tau_s}(u) + \beta(\tau_s)\nabla_u \log \rho_{\tau_s}(u). \quad (5)$$

Since $\rho_t(u) = Z^{-1}\psi(t, u)r(u)$ and $r(u) = \mathcal{N}(u; 0, I_{m_\epsilon})$,

$$\nabla_u \log \rho_t(u) = \nabla_u \log \psi(t, u) - u = g(t, u) - u. \quad (6)$$

Substituting (6) into (5) gives

$$\tilde{b}_s(u) = \beta(\tau_s)g(\tau_s, u) - \tfrac{1}{2}\beta(\tau_s)u,$$

which is exactly the drift in (4). Equivalently, $q_s(u) = \rho_{\tau_s}(u)$ solves its Fokker–Planck equation. By uniqueness in [Assumption A.1](#), the law of $\xi_s^\leftarrow$ has density q_s . The claims at $t_0 = 0$ follow because $\rho_0 = \rho_{\vartheta}(\cdot \mid \mathcal{C})$. The process-space statement follows from [Proposition 2.2](#). $\square$

B Additional numerical experiments and implementation details

In this appendix we provide additional numerical results for the spatiotemporal examples shown in [Figure 2](#), as well as full implementation details.

Each model is implemented as a deterministic map from a Gaussian latent $\mathbf{z}$ to a spatio-temporal field trajectory, enabling the use of LATENTFLOW. All posteriors are obtained by running the integrator sampler for 200 steps with an Euler Maruyama SDE integration and a single Monte Carlo guidance sample per step. All timings are reported on a 13th Gen Intel i9-13900K CPU.

B.1 Advection-diffusion plume

B.1.1 Forward Model

The simulator models a continuously emitting point source on a 22×22 spatial grid with domain $[-1, 1]^2$ over $T = 32$ time steps at horizon $t_{\max} = 2.2$. The full latent vector $\mathbf{z}$ has the following components, each drawn independently from $\mathcal{N}(0, 1)$:

- $z_x^{(s)}, z_y^{(s)}$: control the source location (x_s, y_s) via a tanh squashing, confining the emitter to a small region near $(-0.76, 0)$;
- $z^{(w)}$: sets the mean horizontal wind speed $\bar{u}_x$;
- $z^{(y)}$: sets the base lateral (cross-wind) drift u_y^{base} ;
- $z_0^{(I)}, z_1^{(I)}, \dots, z_{T-1}^{(I)}$: a time-indexed sequence controlling the emission intensity at each step through a cumulative tanh-squashed process $\xi^{(I)}$;

- $z_0^{(w)}, \dots, z_{T-1}^{(w)}$: a time-indexed sequence driving the fluctuating component of the horizontal wind $\xi^{(w)}$;
- two sets of i.i.d. latents driving the slow and fast lateral meander processes ξ^{slow} and ξ^{fast} .

LATENTFLOW performs inference over this entire latent vector, recovering posteriors over source location, wind field, and emission history jointly from observations of the concentration field.

The source location (x_s, y_s) is stochastic, with

$$x_s = -0.76 + 0.12 \tanh(z_x^{(s)}), \quad y_s = 0.45 \tanh(z_y^{(s)}).$$

The concentration field at time t_k is the superposition of all past Gaussian parcels,

$$c(\mathbf{x}, t_k) = \sum_{j=0}^k \frac{\Delta t \cdot \alpha_j \cdot I_j}{2\pi\sigma_j^2} \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_{kj}\|^2}{2\sigma_j^2}\right),$$

where $\sigma_j^2 = \sigma_0^2 + 2\kappa(t_k - t_j)$ with $\sigma_0 = 0.040$ and diffusivity $\kappa = 0.028$; $\alpha_j = e^{-0.28(t_k - t_j)}$ is an age-decay factor; and I_j is a stochastic intensity,

$$\log I_j = 0.28 \xi_j^{(I)} + 0.18 z_0^{(I)}, \quad \xi_j^{(I)} = \tanh\left(0.38 \sum_{i=0}^j z_i^{(I)} \sqrt{\Delta t}\right).$$

The parcel centre for a parcel emitted at t_j , evaluated at t_k , is

$$\mu_{kj}^x = x_s + \bar{u}_x^j(t_k - t_j), \quad \mu_{kj}^y = y_s + u_y^{\text{base}}(t_k - t_j) + 0.52 \Delta \xi_{kj}^{\text{slow}} + 0.18 \Delta \xi_{kj}^{\text{fast}},$$

where $\bar{u}_x^j = 0.72 + 0.08 \tanh(z^{(w)}) + 0.06 \xi_j^{(w)}$ is the horizontal wind, $u_y^{\text{base}} = 0.30 \tanh(z^{(y)})$ is a stochastic base lateral drift, and $\Delta \xi_{kj}^{\text{slow}}, \Delta \xi_{kj}^{\text{fast}}$ are increments of two independent tanh-squashed cumulative Brownian meander processes.

B.1.2 Conditions

We consider three different conditioning regimes (Figure 3). One of these is shown in the main text (Figure 2).

Sparse sensors. A ground-truth trajectory is sampled from the prior. Eight sensor locations are selected as the top-8 highest-concentration pixels in the right half of the domain ($x > 0.20$), averaged over four observation times $t \in \{t_8, t_{14}, t_{20}, t_{26}\}$, giving $8 \times 4 = 32$ scalar observations in total. For each sensor-time pair (i, j) , the condition is an equality constraint,

$$c(\mathbf{x}_i, t_j) = c^*(\mathbf{x}_i, t_j) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 0.02,$$

Obstacle avoidance. The plume must not enter a protected central rectangular region $\mathcal{P} = \{(x, y) : 0.32 < x < 0.82, |y| < 0.34\}$ at any time step. This is a binary condition with smooth margin

$$\phi(c) = 0.06 - \text{smax}_t(\text{smax}_{\mathbf{x} \in \mathcal{P}}[c(\mathbf{x}, t)]) > 0,$$

where smax denotes the log-sum-exp smooth maximum with temperature $\tau = 0.025$.

Detector hit. A small downstream detector region $\mathcal{D} = \{(x, y) : 0.55 < x < 0.90, y < -0.45\}$ must be triggered, i.e., the peak concentration over the detector must exceed a threshold $c^* = 0.40$:

$$\phi(c) = \text{smax}_t(\text{smax}_{\mathbf{x} \in \mathcal{D}}[c(\mathbf{x}, t)]) - 0.40 > 0.$$

This event is rare under the prior, so guidance must steer samples toward trajectories in which the plume reaches the bottom-right corner.

B.2 FitzHugh–Nagumo excitable medium

B.2.1 Forward Model

The simulator integrates the FitzHugh–Nagumo reaction-diffusion system on a 20×20 periodic grid over $T = 34$ steps at horizon $t_{\text{max}} = 4.0$.

The latent vector $\mathbf{z}$ encodes the following independent $\mathcal{N}(0, 1)$ draws:

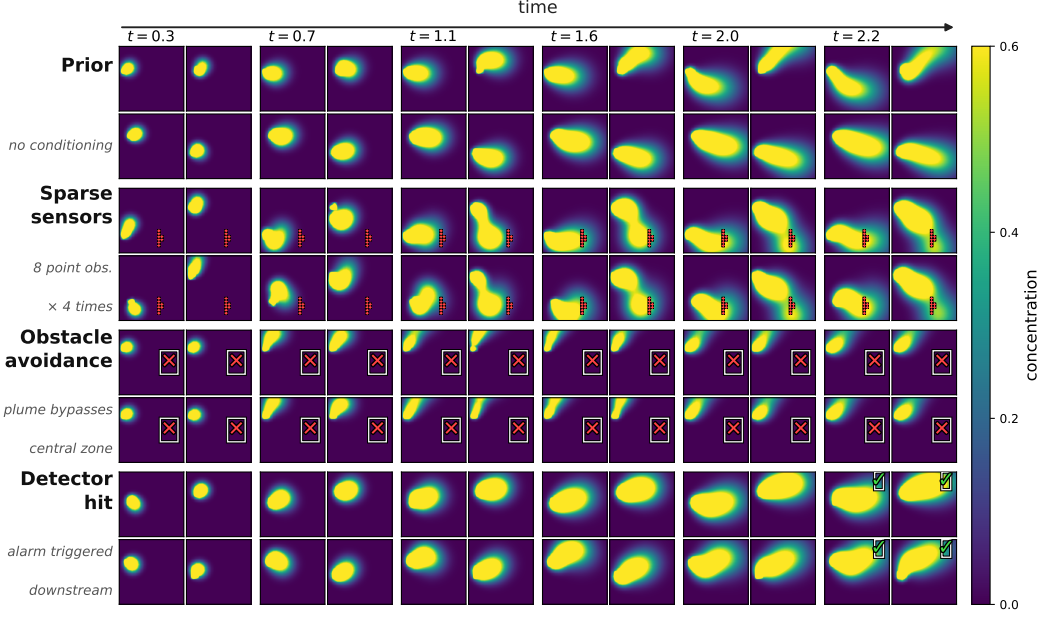


Figure 3: Posterior samples from the advection-diffusion plume model under three conditioning regimes, each showing four independent draws at six time snapshots. *Prior*: unconditioned samples exhibiting diverse plume trajectories and intensities. *Sparse sensors*: conditioning on exact concentration values at 8 spatial locations observed at 4 times (32 observations total) recovers the ground-truth plume trajectory. *Obstacle avoidance*: a binary condition requiring the plume to stay below concentration 0.06 in the central rectangular region (dashed box, $\times$) at all times steers samples around the obstacle. *Detector hit*: a binary condition requiring peak concentration to exceed 0.40 in the downstream detector region (dashed box, $\checkmark$) concentrates mass on trajectories that reach the top-right corner.

- $z_t^{(1)}$: shifts the firing time of pacemaker 1 (upper-left) within the window $[0.27, 0.83]$ via $t_1 = 0.55 + 0.28 \tanh(z_t^{(1)})$;
- $z_t^{(2)}$: shifts the firing time of pacemaker 2 (lower-right) within the window $[1.70, 2.90]$ via $t_2 = 2.30 + 0.60 \tanh(z_t^{(2)})$;
- two additional scalars controlling the amplitudes A_1 and A_2 of each pacemaker;
- $z^{(n)}$: a single scalar that, after box-filter smoothing, produces the fixed spatial heterogeneity field $\eta(\mathbf{x})$ modulating local excitability across the grid.

The SDE noise terms dW_u and dW_v in (7)–(8) are not part of $\mathbf{z}$; they are injected directly during simulation and are not inferred. LATENTFLOW thus performs inference over the low-dimensional latent $\mathbf{z}$ controlling which pacemakers fire, when, and how strongly.

The activator u and recovery variable v evolve as

$$du = \left[D_u \nabla^2 u + u - \frac{u^3}{3} - v + I_{\text{stim}}(\mathbf{x}, t) + \eta(\mathbf{x}) \right] dt + \sigma_u dW_u, \quad (7)$$

$$dv = \epsilon(u + a - bv) dt + \sigma_v dW_v, \quad (8)$$

with $D_u = 0.20$, $\epsilon = 0.095$, $a = 0.72$, $b = 0.82$, $\sigma_u = 0.12$, $\sigma_v = 0.035$. After each Euler–Maruyama step both variables are saturated via $u \leftarrow 2.25 \tanh(u/2.25)$, $v \leftarrow 1.65 \tanh(v/1.65)$. The heterogeneity field $\eta(\mathbf{x}) = 0.11 \mathcal{S}^{(4)}(z^{(n)})$ is a spatially smooth noise field constructed by applying four passes of a box-filter smoother $\mathcal{S}$ to the global latent channel and held fixed over time.

Two spatially localised Gaussian pacemakers provide the stimulus:

$$I_{\text{stim}}(\mathbf{x}, t) = \sum_{i=1}^2 A_i \exp\left(-\frac{(t - t_i)^2}{2 \cdot 0.30^2}\right) \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{2 \cdot 0.17^2}\right),$$

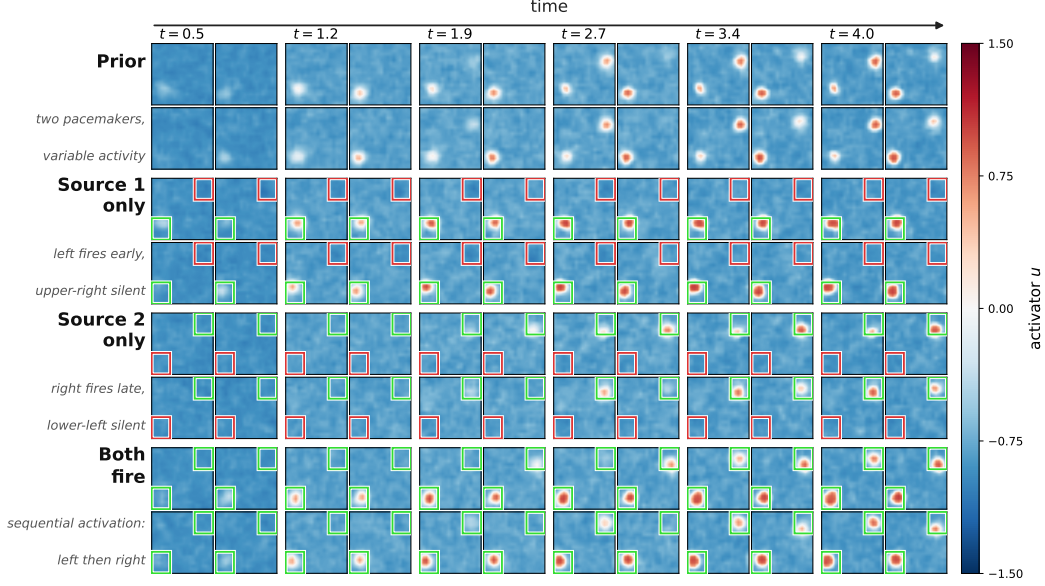


Figure 4: Posterior samples from the FitzHugh–Nagumo excitable medium under three conditioning regimes, each showing four independent draws at six time snapshots. *Prior*: unconditioned samples show variable activation of two spatially separated pacemakers (green boxes: active; red boxes: silent). *Source 1 only*: conditioning on the lower-left pacemaker firing early ($t \lesssim 1.3$) while the upper-right remains silent produces a single travelling wavefront originating from the upper-left corner. *Source 2 only*: the complementary condition suppresses the lower-left source and requires the upper-right to fire late ($t \gtrsim 1.8$), yielding a wavefront from the lower-right. *Both fire*: conditioning on sequential activation of both sources produces two distinct wavefronts propagating across the medium in succession.

with source 1 at $\mathbf{x}_1 \approx (-0.72, 0.55)$ (upper-left) and firing time $t_1 = 0.55 + 0.28 \tanh(z_t^{(1)}) \in [0.27, 0.83]$, and source 2 at $\mathbf{x}_2 \approx (0.68, -0.55)$ (lower-right) and $t_2 = 2.30 + 0.60 \tanh(z_t^{(2)}) \in [1.70, 2.90]$. The amplitudes A_i are also drawn stochastically.

B.2.2 Conditions

We consider three binary conditioning regimes. Let $p(\mathbf{x}, t; \mathcal{M}, \mathcal{T})$ denote the peak activator value over spatial region $\mathcal{M}$ and time window $\mathcal{T}$:

$$p(u; \mathcal{M}, \mathcal{T}) = \max_{t \in \mathcal{T}} \max_{\mathbf{x} \in \mathcal{M}} u(\mathbf{x}, t).$$

Define neighbourhood regions $\mathcal{N}_1 = \{x < -0.45, y > 0.30\}$ and $\mathcal{N}_2 = \{x > 0.45, y < -0.30\}$ for the two sources, and time windows $\mathcal{T}_{\text{early}} = [0, 1.30]$ and $\mathcal{T}_{\text{late}} = [1.80, 4.0]$.

Source 1 only. The lower-left pacemaker fires strongly early while the upper-right stays silent:

$$p(u; \mathcal{N}_1, \mathcal{T}_{\text{early}}) > 0.55 \quad \text{and} \quad p(u; \mathcal{N}_2, \mathcal{T}_{\text{late}}) < -0.75.$$

Source 2 only. The upper-right pacemaker fires late while the lower-left never reaches threshold:

$$p(u; \mathcal{N}_2, \mathcal{T}_{\text{late}}) > 0.55 \quad \text{and} \quad p(u; \mathcal{N}_1, \mathcal{T}_{\text{early}}) < -0.75.$$

Both fire. Sequential activation: lower-left fires early, upper-right fires late:

$$p(u; \mathcal{N}_1, \mathcal{T}_{\text{early}}) > 0.55 \quad \text{and} \quad p(u; \mathcal{N}_2, \mathcal{T}_{\text{late}}) > 0.65.$$

All three are implemented as binary conditions with probit likelihood and $\nu = 10^{-5}$ (10^{-3} for source 2 only), with four Monte Carlo guidance samples per step and gradient norm clipping at 30.0.

B.3 Allen–Cahn phase field

B.3.1 Forward Model

The simulator integrates the stochastic Allen–Cahn equation on a 64×64 periodic grid over $T = 30$ steps with $\Delta t = 1/29$:

$$du = [D\nabla^2 u + r(u - u^3)] dt + \sigma dW,$$

with $D = 2.0$, $r = 1.45$, $\sigma = 0.45$. The double-well potential $f(u) = u - u^3 = -\nabla_u F(u)$ for $F(u) = \frac{1}{4}(1 - u^2)^2$ drives u toward ± 1 , with characteristic interface width $\ell \sim \sqrt{D/r} \approx 1.17$ grid units. The Laplacian uses a nearest-neighbour finite-difference stencil with periodic boundary conditions. After each Euler–Maruyama step the field is saturated via $u \leftarrow 1.45 \tanh(u/1.45)$. The initial condition is

$$u_0(\mathbf{x}) = -0.15 + 0.65 \tanh\left(\mathcal{S}^{(3)}(z_{\mathbf{x}}^{(0)})\right),$$

where $\mathcal{S}^{(3)}$ denotes three passes of a box-filter smoother, which introduces spatial correlations and controls the characteristic length scale of the initial phase domains; $z_{\mathbf{x}}^{(0)} \sim \mathcal{N}(0, I)$ is a 4096-dimensional spatial latent with one scalar per grid point. The SDE noise is likewise reparameterised: at each of the $T = 30$ steps a spatially independent 64×64 Gaussian latent field is drawn and scaled by $\sigma\sqrt{\Delta t}$ to produce the Euler–Maruyama increment σdW . LATENTFLOW performs inference over the full latent vector $\mathbf{z}$ comprising both the initial-condition field and all SDE noise increments.

B.3.2 Conditions

We consider four conditioning regimes are demonstrated. All conditions are imposed on the *final* time step $u(\cdot, t_{29})$ and implemented as equality constraints with observation noise σ_{obs} .

Sparse sensors. A 6×6 regular grid of sensor locations $\{(y_i, x_j)\}$ is placed uniformly across the domain (fractional positions $\{1/12, 3/12, 5/12, 7/12, 9/12, 11/12\}$ in each axis), giving 36 spatial sensors observed at each of 6 snapshot times $t \in \{t_0, t_6, t_{12}, t_{18}, t_{24}, t_{29}\}$, yielding $36 \times 6 = 216$ scalar observations. The condition is

$$u(\mathbf{x}_i, t_k) = u^*(\mathbf{x}_i, t_k) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_{\text{obs}}^2), \quad \sigma_{\text{obs}} = 10^{-5},$$

where u^* is sampled from the prior as a ground truth.

Quadrant means. The spatial domain is divided into four quadrants $\mathcal{Q}_1, \dots, \mathcal{Q}_4$ (upper-left, upper-right, lower-left, lower-right). The condition prescribes the spatial mean of $u(\cdot, t_{29})$ in each quadrant:

$$\frac{1}{|\mathcal{Q}_k|} \sum_{\mathbf{x} \in \mathcal{Q}_k} u(\mathbf{x}, t_{29}) = m_k^*, \quad (m_1^*, m_2^*, m_3^*, m_4^*) = (+0.10, -0.10, -0.10, +0.10),$$

with $\sigma_{\text{obs}} = 10^{-5}$, inducing a diagonal checkerboard mean pattern.

Annular ring. The domain is partitioned into three radially symmetric regions: a core disc of radius $r_{\text{in}} = 0.30$, an annular ring $r_{\text{in}} \leq r \leq r_{\text{out}} = 0.65$, and the exterior $r > r_{\text{out}}$. The condition prescribes the spatial mean in each region at t_{29} :

$$\bar{u}_{\text{ring}} = +0.15, \quad \bar{u}_{\text{core}} = -0.15, \quad \bar{u}_{\text{exterior}} = -0.15,$$

with $\sigma_{\text{obs}} = 10^{-4}$, producing a positive annulus surrounded by negative phases.

Phase front. The domain is divided into three vertical strips by boundaries at $x = \pm 0.30$. The condition prescribes the spatial mean of $u(\cdot, t_{29})$ in each strip:

$$\bar{u}_{\text{left}} = -0.20, \quad \bar{u}_{\text{centre}} = 0.0, \quad \bar{u}_{\text{right}} = +0.20,$$

with $\sigma_{\text{obs}} = 10^{-4}$, inducing a left-to-right phase gradient with a diffuse interface in the central strip.

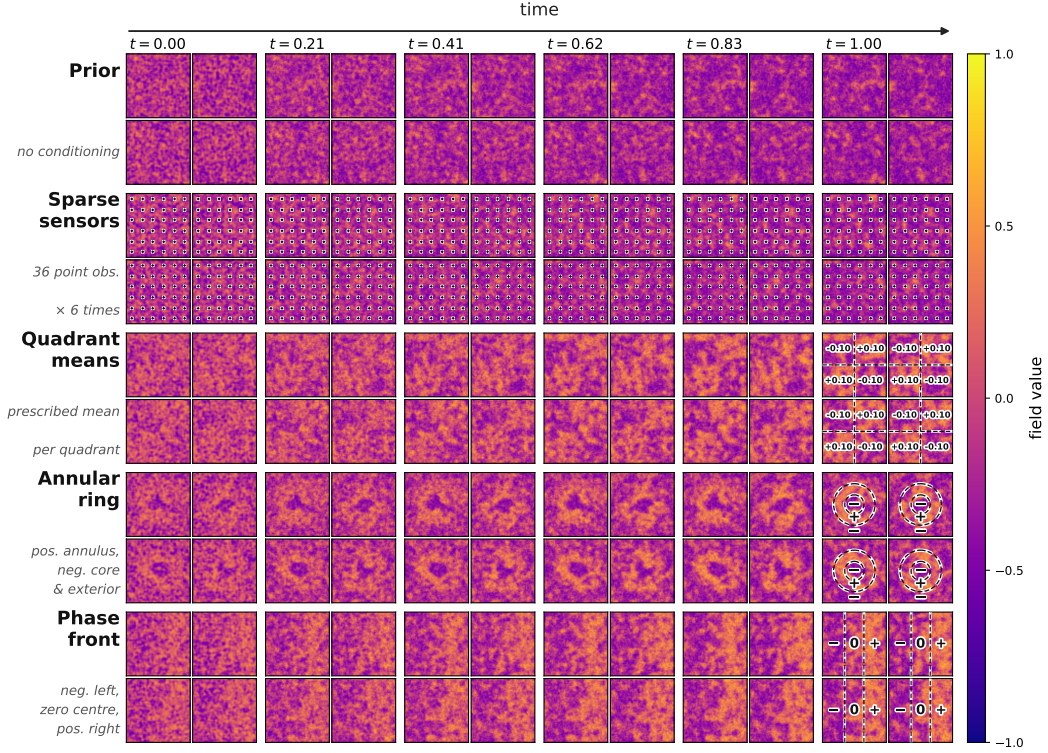


Figure 5: Posterior samples from the Allen–Cahn phase-field model under four conditioning regimes, each showing four independent draws at six time snapshots. All conditions constrain statistics of the final field $u(\cdot, t_{29})$. *Prior*: unconditioned samples exhibit stochastic domain coarsening toward the ± 1 phases. *Sparse sensors*: conditioning on 36 point observations at each of 6 snapshot times (216 observations total) recovers the ground-truth field trajectory. *Quadrant means*: prescribing the spatial mean in each quadrant to a checkerboard pattern $(+0.10, -0.10, -0.10, +0.10)$ steers the final phase arrangement into the corresponding diagonal structure. *Annular ring*: prescribing a positive mean in an annular band ($0.30 \leq r \leq 0.65$) and negative means in the core and exterior produces a ring-shaped positive domain surrounded by negative phase. *Phase front*: prescribing a left-to-right mean gradient $(-0.20, 0.0, +0.20)$ across three vertical strips ($|x| \leq 0.30$ boundaries) induces a diffuse phase interface running vertically through the domain.